Team Name: Bio Lords

Project Title: From Sequence to Subcellar: Deep Learning Architectures for Protein Localization

Project summary:

The human cell has thousands of macromolecules that serve a myriad of functions. The macromolecules, more commonly referred to as proteins, aid in DNA replication, apoptosis (cell death), structural support, call communication, and much more. Depending on the function of a particular protein, it resides in a specific part of a cell. For example, proteins in DNA replication reside near the center while those involved in cell communication can reside near the cell surface. Several factors determine the subcellular location of a protein; however, the most crucial part is the protein amino acid sequence (the building blocks of proteins).

The purpose of this project is to architect and train deep neural network(s) to predict the subcellular location of a protein given its amino acid sequence. The impact of accomplishing this task has heavy implication in various biological facets like drug discovery and disease diagnosis. Understanding protein localization can aid in designing drug delivery systems that target specific cellular compartments. Furthermore, in certain diseases like cancer certain proteins misbehave and even mis-localize, thus characterizing a proteins localization assists in the creation of more effective chemotherapy.

Approach:

After some initial research, there as many implementations to address the issue of predicting a proteins subcellular location. As an amino acid sequence for various proteins can vary in length most studies use a sequence-based model, which we plan to echo as well. More specifically we are interested in learning more about the ability of an attention/transformer-based architecture in this task.

In addition to initial research that has been conducted, the team will attempt to experiment with different deep learning architectures. Such architectures shall include, transformer-based models, recurrent neural networks (RNN), and convolutional neural networks (CNN). After experimenting with different architectures, the team will compare these approaches to understand strengths, weaknesses, and tradeoffs when it comes to determining subcellular location based on protein sequences. As seen within research papers on this topic, there are already reports and metrics that demonstrate a variation of these models and the team will use these reports as a platform to help guide the team's models success.

We would extract comprehensive datasets from UniProt, Human Protein Atlas, and other protein databases, preprocess the data to remove redundancy and ensure data quality, create balanced training sets for major subcellular compartments, and split data into training, validation, and testing sets with consideration for sequence homology.

Following this, we would explore and compare several architectural approaches such as transformer-based models, bidirectional LSTM/GRU layers for capturing long-range dependencies and attention mechanisms for identifying critical sequence regions.

Resources/Related Work:

The task of predicting a protein's subcellular location given a protein sequence has been extensively studied via machine learning and deep learning techniques. Several algorithms (i.e. transformer, RNN, CNN, etc.) have been developed to classify proteins into specific localization sites, with examples such as DeepLoc performing well in this space. Today, there are multiple research papers and open-source projects via GitHub that the team plans on using as a guideline on comparing performances. The team aims to build upon existing research in developing and experimenting with other deep learning models. The team will then iterate and fine tune these models to improve the models' accuracy when it comes to predicting protein subcellular localization.

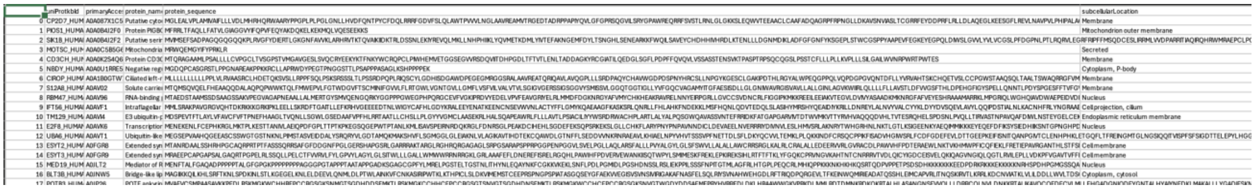
- https://academic.oup.com/nar/article/50/W1/W228/6576357
- https://pmc.ncbi.nlm.nih.gov/articles/PMC11066471/
- https://www.sciencedirect.com/science/article/pii/S0022283600939032
- https://pubmed.ncbi.nlm.nih.gov/15647269/
- https://github.com/uhlerlab/PUPS/tree/main
- https://github.com/JJAlmagro/subcellular_localization/tree/master
- https://pmc.ncbi.nlm.nih.gov/articles/PMC5187447/

Datasets:

Uniprot: https://www.uniprot.org/uniprotkb?query=*

250 million + records of protein sequence & cell location (this covers a variety of species, but the Human proteins are about 500k)

A small subset of the data is shown below:



	uniProtkbId	primaryAcce	protein_name	protein_sequence	subcellularLocation
0	CP2D7_HUM	A0A087X1C5	Putative cyto	[illegible]	Membrane
1	PIOS1_HUMA	A0A0B4J2F0	Protein PIGB	MFRRLTFAQLLFATVLGIAGGVYIFQPVFEQYAKDQKELKEKMQLVQESEEKKS	Mitochondrion outer membrane
2	SIK1B_HUMA	A0A0B4J2F2	Putative serir	[illegible]	
3	MOTSC_HUM	A0A0C5B5G6	Mitochondria	MRWQEMGYIFYPRKLR	Secreted
4	CD3CH_HUM	A0A0K2S4Q6	Protein CD3C	[illegible]	Membrane
5	NBDY_HUMA	A0A0U1RRE5	Negative reg	MGDQPCASGRSTLPPGNAREAKPPKKRCLLAPRWDYPEGTPNGGSTTLPSAPPPASAGLKSHPPPPEK	Cytoplasm, P-body
6	CIROP_HUM	A0A1B0GTW	Ciliated left-r	[illegible]	Membrane
7	S12A8_HUM	A0AV02	Solute carrie	[illegible]	Membrane
8	RBM47_HUM	A0AV96	RNA-binding	[illegible]	Nucleus
9	IFT56_HUMA	A0AVF1	Intraflagellar	[illegible]	Cellprojection, cilium
10	TM129_HUM	A0AVI4	E3 ubiquitin-p	[illegible]	Endoplasmic reticulum membrane
11	E2F8_HUMAN	A0AVK6	Transcription	[illegible]	Nucleus
12	UBA6_HUMA	A0AVT1	Ubiquitin-like	[illegible]	
13	ESYT2_HUM	A0FGR8	Extended syn	[illegible]	Cellmembrane
14	ESYT3_HUM	A0FGR9	Extended syn	[illegible]	Cellmembrane
15	MED19_HUM	A0JLT2	Mediator of R	[illegible]	Nucleus
16	BLT3B_HUM	A0JNW5	Bridge-like lip	[illegible]	Cytoplasm, cytosol
17	POTB3_HUM	A0JP26	POTE ankyrin	[illegible]	

Note that there are other datasets that other papers use that we could as well:

https://services.healthtech.dtu.dk/services/DeepLoc-2.0/ - Used by the DeepLoc paper

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